



Embedding innovation: How SmartDFE transforms FUJIFILM's integrated printbar systems

A new purpose-built system pushes the innovation envelope for FUJIFILM Integrated Inkjet Solutions.

In many industrial processing environments, a standalone digital production printing press may not be a suitable solution. That's where FUJIFILM Integrated Inkjet Solutions comes in. With its smaller, customized inkjet systems, FUJIFILM enables printing to be embedded directly into manufacturing processes, bringing the benefits of inkjet to places where a full production printing press would not fit or be appropriate.

"There are a lot of benefits to inkjet printing, but a standalone production printing press isn't always practical," explains Ankur Mour, Vice President and General Manager of FUJIFILM Integrated Inkjet Solutions. "Our imprinting solutions extend beyond traditional setups and let customers do more with their existing production lines."

FUJIFILM's modular imprint systems, such as the FUJIFILM 12K and 42K Printbar Systems, give businesses flexibility and the ability to run more efficiently. Powered by FUJIFILM SAMBA® printhead technology, these systems deliver high-speed, high-quality productivity.

This in turn enables just-in-time manufacturing, on-demand customization, and production efficiencies for a range of applications in diverse industries.

"There's simply no 'typical' application," continues Mour. "Customers come to us when they want something unusual that we can address with our technology. We then quickly understand their needs and adapt to them with a suitable integrated solution."

Many systems are integrated into workflows for packaging, such as folding cartons and corrugated printing. Other applications include commercial, transactional, trans-promo, and direct-mail printing. These applications require exceptional reliability and precision. In direct mail or packaging applications, customers typically keep analogue processes for static content. Digital imprinting for variable data is added as text, logos, images, QR codes, and bar codes. In a transactional workflow it may involve producing a personalized financial statement, an introductory letter, and a color booklet, all collated into a customized envelope, with the strict requirement that

every element matches the intended recipient. Beyond accuracy, compliance and data protection requirements add another layer of complexity; sensitive data must be handled securely, with zero tolerance for error.

Built on Fundamentals

Originally, FUJIFILM's modular printbar systems were powered by Fundamentals™, a suite of 'off-the-shelf' components developed by Hybrid Software Helix, formerly Global Graphics Software, with Hybrid Software. Fundamentals added workflow features and file rendering with trusted tools including CLOUDFLOW and the Harlequin RIP® raster image processor.

The solutions were popular from the start. "What began as a few projects quickly became many," says Mour. "As demand grew, the market wanted higher performance, more features, more applications, and greater reliability. We kept evolving the systems and so we needed software and electronics that could evolve with us."

“SmartDFE gives us the ability to meet those demands with confidence. It helps us deliver stronger, more user-friendly solutions, while also preparing our customers for the next wave of automation and digital transformation.”

Ankur Mour, Vice President and General Manager, FUJIFILM Integrated Inkjet Solutions.

Enter SmartDFE

It became apparent that a new innovative solution from Hybrid Software Helix, SmartDFE™ for Industrial, an AI-powered digital front end platform, was a more appropriate fit. SmartDFE is designed specifically for integrators and industrial environments to drive printheads like the FUJIFILM SAMBA. It is purpose built for industrial inkjet printing so it deploys faster, integrates seamlessly into existing systems, is more flexible to support different types of digital solutions - and it fulfills some crucial market requirements:

Reliability and uptime - Commercial printers, and especially packaging converters, can't afford downtime. SmartDFE's AI-powered patented Autotune™ feature



The FUJIFILM 12K Printbar System supports commercial printing, direct mail, and transaction documents as well as packaging applications such as folding cartons and corrugated printing.

analyzes each PDF and configures the system for optimal performance, minimizing underruns and keeping the press running at full-rated speed.

Scalability and future-proofing - The Harlequin Direct™ RIP in SmartDFE is easy to scale. If a customer needs more performance, additional RIP servers can be added instantly, distributing color channels across servers for faster throughput.

Connectivity and automation - Built for Industry 4.0, SmartDFE includes embedded OPC UA connectivity to connect seamlessly with factory systems and cloud platforms, ensuring automated, integrated workflows.

The best in color management - Color accuracy is guaranteed with embedded ColorLogic® color management for precise spot colors.

Smart job cost estimation - Supports sales people by accurately estimating the cost of ink, substrate and services for any job.

Customization through modular plug-ins - SmartDFE supports custom plug-ins tailored to individual customer needs. This modular approach adds functionality with minimal disruption. This can be added easily using a print SDK such as the Mako Core™ from Helix.

A strong partnership and good support

"Collaboration has been key," continues Mour. "Our relationship with Hybrid Software Helix is very important. We work closely with their team to fulfill our customers' needs quickly. Support is also easy with the simple support package function within SmartDFE.

"As our customers see new opportunities, they naturally demand more from us," adds Mour. "SmartDFE gives us the ability to meet those demands with confidence. It helps us deliver stronger, more user-friendly solutions, while also preparing our customers for the next wave of automation and digital transformation.

"Our long-standing collaboration with Hybrid Software Helix has already delivered successful systems, but what makes this partnership truly valuable is how we've been able to turn lessons learned into innovations. Together, we've developed solutions that not only build on past success but also exceed our expectations for performance and reliability."



The compact, high-speed FUJIFILM Mini 4300 Printbar System for addressing, variable data images, labels and serialization.

About FUJIFILM Integrated Inkjet Solutions

FUJIFILM Integrated Inkjet Solutions is a business unit within FUJIFILM focused on integrating inkjet technology. It provides both the inkjet technology and the expertise to integrate inkjet into existing or new production lines. Solutions include printheads, inks, and the systems needed to implement them, often for commercial printing, packaging, and other industrial sectors.

inkjet-integration.fujifilm.com



hybridhelix.com/smartdfe

Hybrid Software Inc.
One South State Street
Newtown, PA18940
United States of America
Tel: +1(215) 550-6243

Hybrid Software Helix Ltd
Building 2030
Cambourne Business Park
Cambourne, Cambridge
CB23 6DW UK
Tel: +44 (0)1954 283100

Global Graphics KK
613 AIOS Nagatacho Bldg.
2-17-17 Nagatacho, Chiyoda-ku,
Tokyo 100-0014
Japan
Tel: +81-3-6273-3198

